

Generation Interconnection System Impact Study Report

For Queue Project Q16-0573

Jennings Lodge 100 kV

November 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0573. The Interconnection Customer proposes a 155.4 MW solar photovoltaic facility interconnecting at the Jennings Lodge 100 kV substation in the Indiana area. The request was received on January 14, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 218 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$594,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Big Oaks - Cypress Creek 100 KV
Contingency: Sate St - Robison Park 230 KV Line
Loading: 125.7% (Rating: 241 MVA)
DFAX: 5.7%
Pre-project loading: 122.0%
2. Facility: Ligonier - Cypress Crk 100 KV
Contingency: State Street - Robison Park 230 KV
Loading: 116.4% (Rating: 269 MVA)
DFAX: 5.7%
Pre-project loading: 113.2%
3. Facility: Big Oaks - Laureles 100 KV
Contingency: Robison Park - State St 230 KV
Loading: 105.3% (Rating: 265 MVA)
DFAX: 5.7%
Pre-project loading: 102.0%
4. Facility: Golden Gate - Laotto 100 KV Line
Contingency: State St - Robison Park 230 KV
Loading: 101.7% (Rating: 193 MVA)
DFAX: 5.7%
Pre-project loading: 97.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Oaks - Cypress Creek 100 KV	Rebuild line with higher-capacity conductor	\$4,840,000	4% (\$193,000)
Ligonier - Cypress Crk 100 KV	Rebuild line with higher-capacity conductor	\$5,793,000	4% (\$231,000)
Big Oaks - Laureles 100 KV	Reconductor line	\$883,000	4% (\$35,000)
Golden Gate - Laotto 100 KV	Reconductor line	\$2,325,000	6% (\$135,000)
	Total allocated to project		\$594,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.